

Project Requirements Document (PRD)

- Version 3.0 (Sensor Fusion Update)

Project Name: Edge-AI Fall Detection & Intelligent Emergency Response System

Primary Technologies: Nvidia Jetson Nano, Python, Computer Vision (CV), WebRTC, Speech-to-Text (STT), Webhooks, BLE (Bluetooth Low Energy), Wearable Sensors



1. Project Overview

This system utilizes an Nvidia Jetson Nano to perform real-time fall detection at the edge, leveraging a multi-modal approach. By combining Computer Vision (CV) with a custom wearable wristband (equipped with an accelerometer and heart rate sensor), the system utilizes **sensor fusion** to achieve highly accurate fall detection. Upon detecting an anomaly, the system uses interactive voice prompts to verify the subject's status. If an emergency is confirmed—either verbally, through silence, or critical vital signs—it escalates the situation by utilizing WebRTC for live audio-visual feeds and Webhooks to instantly alert caregivers and emergency responders.

2. Key Functional Requirements

2.1. Vision & Wearable Sensor Fusion (New Integration)

- **Wearable Wristband:** The subject will wear a custom wristband equipped with an Accelerometer (IMU) and a Photoplethysmography (PPG) Heart Rate sensor.
- **Real-Time Telemetry:** The wristband will continuously transmit motion data and heart rate metrics to the Jetson Nano via Bluetooth Low Energy (BLE) or local Wi-Fi.
- **Multi-Modal Detection Logic:**
 - * **Accelerometer Data:** Used to detect sudden impacts or free-fall signatures, acting as a critical secondary verification to the CV's visual pose estimation. This drastically reduces false positives (e.g., the CV dropping a bounding box).
 - **Heart Rate Monitoring:** Continuously monitors for abnormal spikes (panic/shock) or dangerous drops following a detected fall, providing critical context to the severity of the event.

2.2. Intelligent Fall Verification & Voice Interaction

- **Status Check Routine:** When a fall is flagged by the CV and/or Accelerometer, the system outputs an automated voice prompt (e.g., *"A fall has been detected. Are you okay?"*).
- **Audio Capture & Analysis:** The microphone listens for a response using local Speech-to-Text (STT).
 - **Emergency Escalation Criteria:**
 1. Verbal distress (e.g., "Help," "Emergency").
 2. Complete silence for a predefined timeout (e.g., 10 seconds).
 3. **[New]** Critical heart rate readings detected by the wristband during the verification phase, regardless of verbal response.
 - **False Alarm Mitigation:** Safe keywords cancel the alert if vital signs are stable.

2.3. WebRTC & Webhook Alert System

- **WebRTC Live Feed:** Automatically establishes a secure, peer-to-peer audio/video connection to a caregiver's dashboard for real-time visual assessment.
- **External Webhook Dispatch:** Triggers pre-configured Webhooks (SMS, Slack, 119/911 APIs) to send automated messages containing:
 - Subject ID and Location.
 - Nature of alert (e.g., "Unresponsive Fall").
 - **[New]** Current vital signs (Heart rate at the time of the fall).

3. Updated System Architecture & Logic Flow

3.1. Hardware Components

- **Processing Edge Node:** Nvidia Jetson Nano.

- **Sensory Input/Output (Local):** Webcam (Vision), Microphone (Audio In), Speaker (Audio Out).
- **Wearable Device (Remote Local):** Custom Wristband (Accelerometer + Heart Rate Sensor, BLE/Wi-Fi Microcontroller like ESP32).
- **Network Interface:** Wi-Fi/Ethernet for WebRTC/Webhooks, Bluetooth module for sensor communication.

3.2. Automated Response Flow

1. **Continuous Monitoring:** Jetson Nano processes webcam feed (CV) *and* listens to continuous BLE data streams (Heart Rate + Accelerometer) from the wristband.
2. **Detection & Sensor Fusion:** A physical fall is flagged if the CV algorithm detects a rapid posture change AND/OR the accelerometer registers an impact spike.
3. **Verification:** System plays audio prompt: *"Are you okay?"* and begins listening.
4. **Judgment:** Evaluates STT audio response against real-time heart rate data.
 - *Positive Response + Stable Vitals:* Alert canceled.
 - *Negative Response OR Silence OR Critical Vitals:* Active emergency flagged.
5. **Action:** Fires Webhooks with location and vital sign data, and opens WebRTC channels for live visual monitoring by caregivers.

4. Non-Functional Requirements

- **Connectivity Reliability:** The Jetson Nano must feature an auto-reconnect logic for the BLE/Wi-Fi connection to the wristband in case of signal drops.
- **Power Efficiency (Wearable):** The wristband's firmware must be optimized to ensure a battery life of at least 24-48 hours to minimize charging frequency for the user.
- **Data Synchronization:** Telemetry from the wristband and video frames from the webcam must be tightly time-synced on the Jetson Nano to ensure the sensor fusion algorithms evaluate the exact same moment in time.

5. Sensor Fusion Algorithm & Emergency Classification Logic

The Jetson Nano will utilize a rule-based logic framework to continuously evaluate the synchronized data streams from the Computer Vision (CV) model, Accelerometer, and Heart Rate sensor. This multi-modal approach classifies the event into different severity tiers to dictate the appropriate response.

5.1. Scenario A: Standard Fall (Physical Trauma / Conscious)

- **Data Signatures:**
 - **CV:** Rapid downward translation of the bounding box.
 - **Accelerometer:** A single, distinct high-G impact peak followed by stillness.
 - **Heart Rate:** A sudden, moderate spike (e.g., 100–120 BPM) caused by the "fight or flight" adrenaline response, surprise, or pain.

- **System Action (Tier 1 - Verification):** Triggers the standard voice interaction protocol. The speaker asks, "Are you okay?" and waits for a verbal response via STT to determine if WebRTC/Webhooks should be activated.

5.2. Scenario B: Suspected Seizure & Fall (Ictal Tachycardia)

- **Data Signatures:**
 - **CV:** Subject detected in a prone/fallen position.
 - **Accelerometer:** Initial impact peak, immediately followed by **continuous, rhythmic, low-amplitude oscillations** (indicating muscle tremors/convulsions).
 - **Heart Rate:** An instantaneous, extreme spike (e.g., >130–160 BPM) occurring simultaneously with the fall, characteristic of seizure-induced autonomic surges.
- **System Action (Tier 2 - Critical Emergency):** Bypasses the verbal verification loop entirely. The system assumes the subject is incapacitated and immediately triggers Webhooks to emergency services (noting suspected seizure) and opens the WebRTC live feed for the caregiver.

5.3. Scenario C: Syncope or Cardiovascular Event (Fainting)

- **Data Signatures:**
 - **CV / Accelerometer:** Posture collapse or impact detected.
 - **Heart Rate:** A severe, sudden drop (Bradycardia, e.g., <50 BPM) prior to or during the fall, or a complete loss of pulse signal (potential cardiac arrest).
- **System Action (Tier 2 - Critical Emergency):** Bypasses verbal verification. Instantly flags a life-threatening medical emergency. Dispatches Webhooks with critical vitals alerts and initializes the WebRTC stream.

5.4. Scenario D: False Positive Mitigation (Dropped Object / Bending Down)

- **Data Signatures:**
 - **CV:** Bounding box lowers, but quickly returns to a standing position, OR CV drops tracking temporarily.
 - **Accelerometer:** No significant impact peak detected.
 - **Heart Rate:** Remains stable at the user's baseline resting rate.
- **System Action (Tier 0 - Dismissal):** The system cross-references the lack of physical impact and stable vitals to override the CV anomaly. No alerts or voice prompts are triggered; standard monitoring resumes.